Supplementary Material: Trace-Aware Power System Scheduling with Heterogeneous Power Sources: A Learning-Based Framework

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This document accompanies the paper cited above. It describes, in a step-by-step manner, the test-system settings, the neural-network training procedure, the exact mixed-integer linear reformulation of the trained network, and the supplementary figures that support the numerical studies. All quantities are reported in the same units as in the main paper, and the notation is consistent with the one used there.

S1. Modified IEEE 14-Bus Test System

The small-system study is carried out on a modified IEEE 14-bus system. The system data are taken from the MATPOWER `case14` test case, and branch 4–5 is removed so that the wind source at Bus 3 and the renewable-share tracing requirement at Bus 4 are separated by the network in a well-defined way. The resulting system has 14 buses, 19 branches, and five thermal units located at Buses 1, 2, 3, 6, and 8. A 100 MW wind farm is connected at Bus 3, and the tracing requirement is imposed at Bus 4. The scheduling horizon is 24 hours with an hourly resolution.

Table S1 summarizes the principal system and scheduling settings. The base power is 100 MVA, which is also used as the per-unit base in the network model. The minimum output of every thermal unit is $0.2P^{\max}$, the ramp-up and ramp-down limits are $0.6P^{\max}$ per hour, and the minimum up and down times are both 3 hours. The startup cost is \$3000 per start. The quadratic fuel-cost curves are replaced by a four-segment linear approximation, so the resulting scheduling model is a mixed-integer linear program whose accuracy can be controlled by the number of segments.

Table S2 lists the parameters of the five thermal units. Unit TU1 at Bus 1 is the largest unit, with a maximum output of 332.4 MW, and the remaining four units have maximum outputs of 140 MW or 100 MW. The minimum output of each unit is 20% of its maximum output, which is consistent with the general setting in Table S1. The quadratic cost of each unit is $c_2P^2 + c_1P + c_0$, where P is the power output in MW; c_2 is in \$/MW^2, c_1 in \$/MWh, and c_0 in \$. The four-segment approximation is built from these coefficients for every unit.

Table S3 summarizes the neural-network surrogate used in this case. The training data are generated from historical wind and dispatch traces; after preprocessing, 34,427 samples are retained and split into training, validation, and test sets with the ratio 70%/15%/15%. The input vector contains the total wind power and the outputs of TU1–TU5, and the outputs are the renewable-share tracing variables at Buses 3 and 4. The network has two hidden layers with 24 and 12 neurons, respectively, and ReLU activations.

TABLE S1
PRINCIPAL SYSTEM AND SCHEDULING SETTINGS OF THE MODIFIED IEEE 14-BUS TEST SYSTEM.

Parameter	Value
Base power	100 MVA
Buses / branches	14 / 19
Thermal units	5
Thermal-unit buses	1, 2, 3, 6, 8
Wind farm	100 MW at Bus 3
Requirement bus	Bus 4
Horizon / interval	24 h / 1 h
Minimum output	$0.2P^{\max}$
Ramp-up/down limits	$0.6P^{\max}$/h
Minimum up/down time	3 h / 3 h
Startup cost	\$3000/start
Cost approximation	Four segments

TABLE S2
THERMAL-UNIT PARAMETERS OF THE MODIFIED IEEE 14-BUS TEST SYSTEM.

Unit	Bus	P (MW)		c_2	c_1	c_0
		$P^{\max}$	$P^{\min}$			
TU1	1	332.4	66.48	0.0430	19	0
TU2	2	140.0	28.00	0.2500	20	0
TU3	3	100.0	20.00	0.0100	40	0
TU4	6	100.0	20.00	0.0100	41	0
TU5	8	100.0	20.00	0.0100	42	0

TABLE S3
NEURAL-NETWORK SETTINGS OF THE MODIFIED IEEE 14-BUS TEST SYSTEM.

Parameter	Value
Retained samples	34,427
Train / validation / test	70% / 15% / 15%
Inputs	Total wind; TU1–TU5 outputs
Outputs	Shares at Buses 3 and 4
Hidden layers	[24, 12]
Activation	ReLU

This document is the supplementary material accompanying the paper "Trace-Aware Power System Scheduling with Heterogeneous Power Sources: A Learning-Based Framework." (Corresponding author: Kai Xiong.)

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TABLE S4
HOURLY LOAD AND WIND FACTORS USED IN THE 24-H SCHEDULING STUDY.

Hour	Load	Wind	Hour	Load	Wind
1	0.7389	0.6602	13	0.8923	0.6053
2	0.7187	0.6830	14	0.8877	0.6038
3	0.7035	0.7083	15	0.8840	0.6068
4	0.6983	0.6925	16	0.8849	0.9050
5	0.7040	0.6845	17	0.8941	0.8218
6	0.7251	0.7142	18	0.9345	0.8220
7	0.7764	0.7235	19	0.9899	0.6870
8	0.8461	0.5415	20	0.9841	0.6559
9	0.8951	0.7793	21	0.9691	0.6720
10	0.9094	0.7860	22	0.9436	0.6934
11	0.9078	0.7215	23	0.9094	0.7058
12	0.9071	0.6585	24	0.8674	0.6964

The base active-load vector of the modified system is

$$P^{\mathrm{L}}_{\mathrm{base}} = [0, 21.7, 94.2, 47.8, 7.6, 11.2, 0, 0, 29.5, 9.0, 3.5, 6.1, 13.5, 14.9]\ \mathrm{MW}, \tag{S1}$$

where the entries correspond to Buses 1 to 14 in order. The load at bus n and hour t is obtained by multiplying the corresponding entry by the hourly load factor reported in Table S4; the wind power at hour t is likewise obtained by multiplying the rated wind capacity by the hourly wind factor. Both profiles are listed for all 24 hours.

S2. NEURAL-NETWORK TRAINING AND MILP REFORMULATION

This section documents how the neural-network surrogate is trained and how the trained network is embedded into the scheduling model as exact mixed-integer linear constraints. The notation is the same as in the paper.

a) Normalization.: For sample m, let $\mathbf{x}^{(m)}$ and $\mathbf{T}^{\mathrm{G},(m)}$ denote the input and tracing-label vectors, and let $(\boldsymbol{\mu}_x, \boldsymbol{\sigma}_x)$ and $(\boldsymbol{\mu}_y, \boldsymbol{\sigma}_y)$ be the corresponding sample means and standard deviations. The operator $\oslash$ denotes elementwise division. Each input and output is normalized to zero mean and unit variance before training:

$$\bar{\mathbf{x}}^{(m)} = (\mathbf{x}^{(m)} - \boldsymbol{\mu}_x) \oslash \boldsymbol{\sigma}_x, \tag{S2a}$$

$$\bar{\mathbf{y}}^{(m)} = (\mathbf{T}^{\mathrm{G},(m)} - \boldsymbol{\mu}_y) \oslash \boldsymbol{\sigma}_y. \tag{S2b}$$

Normalization prevents features with different physical scales, such as megawatt-level power and dimensionless shares, from dominating the training objective.

b) Forward pass and training loss.: The function $\sigma(\cdot)$ is the ReLU activation, and Θ collects the trainable weights and biases. The forward pass of the fully connected network with L hidden layers is

$$\mathbf{z}^{1,(m)} = \sigma(\mathbf{W}^1 \bar{\mathbf{x}}^{(m)} + \mathbf{b}^1), \tag{S3a}$$

$$\mathbf{z}^{p,(m)} = \sigma(\mathbf{W}^p \mathbf{z}^{p-1,(m)} + \mathbf{b}^p), \quad p = 2, \ldots, L, \tag{S3b}$$

$$\hat{\mathbf{y}}^{(m)} = \mathbf{W}^{L+1} \mathbf{z}^{L,(m)} + \mathbf{b}^{L+1}, \qquad \Theta = \{\mathbf{W}^p, \mathbf{b}^p\}_{p=1}^{L+1}, \tag{S3c}$$

where $\hat{\mathbf{y}}^{(m)}$ is the network output in the normalized space. Training minimizes the mean squared error between the normalized prediction and the normalized label, and the trained output is mapped back to the physical space by elementwise multiplication ($\odot$) with the output standard deviation followed by addition of the output mean:

$$\min_{\Theta} \frac{1}{N_{\mathrm{sam}}} \sum_{m=1}^{N_{\mathrm{sam}}} \|\hat{\mathbf{y}}^{(m)} - \bar{\mathbf{y}}^{(m)}\|_2^2, \tag{S4a}$$

$$\widetilde{\mathbf{T}}^{\mathrm{G},(m)} = \boldsymbol{\mu}_y + \boldsymbol{\sigma}_y \odot \hat{\mathbf{y}}^{(m)}. \tag{S4b}$$

In the scheduling model, the same forward mapping is evaluated period by period by replacing the sample index m with the scheduling index t; the normalization statistics are computed from the training data and kept fixed during scheduling.

c) Exact MILP embedding of the ReLU network.: For a hidden neuron with affine input $\hat{z}$, ReLU output $z = \max\{\hat{z}, 0\}$, and valid bounds $\underline{h} \le \hat{z} \le \bar{h}$, introduce a binary activation variable a. Its exact mixed-integer linear representation is

$$z \ge 0, \qquad z \ge \hat{z}, \tag{S5a}$$

$$z \le \hat{z} - \underline{h}(1-a), \qquad z \le \bar{h} a, \qquad a \in \{0,1\}. \tag{S5b}$$

When $a = 0$, the constraints force $z = 0$ and require $\hat{z} \le 0$; when $a = 1$, they give $z = \hat{z} \ge 0$. Hence the binary variable selects exactly the active or inactive branch of the ReLU. The bounds $\underline{h}$ and $\bar{h}$ are propagated through the network by interval arithmetic, so the big-M coefficients are finite and tight. Applying these constraints neuron by neuron, followed by the affine output layer and the denormalization step in (S4), yields the exact MILP embedding used in the paper. Because the embedding is exact, the approximation in the scheduling model comes from the piecewise-linear representation of the fuel-cost curves.

S3. PRACTICAL-SCALE TRANSMISSION-SYSTEM CASE

To demonstrate the scalability of the proposed approach, a larger transmission-system case is constructed from the MATPOWER `case57` test case. The system has 57 buses and 78 branches. The 50 thermal units of the original case are aggregated at 16 buses, and eight wind farms of 1500 MW each are connected at Buses 7, 13, 23, 25, 42, 47, 49, and 53, giving a total wind capacity of 12,000 MW. The renewable-share tracing requirement is imposed at Bus 55, and Bus 38 is chosen as the slack bus. The line ratings of the original case are multiplied by 3 so that the transmission limits do not dominate the dispatch results and the focus remains on the scheduling and tracing decisions.

Table S5 summarizes the principal settings of this case.

Table S6 reports the aggregated thermal-unit parameters by bus. For each bus, the table gives the number of units connected to that bus and the totals of their maximum output, minimum output, ramp-up/ramp-down capability (RU/RD), and linear cost coefficient c_1. The listed capacity and ramping values are therefore totals over the units at the corresponding bus. The quadratic and constant cost coefficients are common to all units, with $c_2 = 0.01$ and $c_0 = 0$. As in the small system, the quadratic costs are represented by the four-segment linear approximation reported in Table S5, and the scheduling model is again solved as a mixed-integer linear program.

TABLE S5
PRINCIPAL PRACTICAL-SCALE CASE SETTINGS.

Parameter	Value
Base power	100 MVA
Buses / branches	57 / 78
Thermal units	50
Wind farms	8 × 1500 MW
Wind-farm buses	7, 13, 23, 25, 42, 47, 49, 53
Total wind capacity	12,000 MW
Requirement / slack bus	55 / 38
Horizon / interval	24 h / 1 h
Line-rating multiplier	3
Cost approximation	Four segments

TABLE S6
THERMAL-UNIT PARAMETERS BY BUS IN THE PRACTICAL-SCALE CASE.

Bus	Units	$P^{\max}$ (MW or MW/h)	$P^{\min}$	RU/RD	c_1
3	2	1200	240	720	38.5000
8	2	1320	264	792	41.5000
9	4	1200	240	720	20.0000
11	6	4400	880	2640	30.9091
14	2	1200	240	720	40.0000
24	4	2020	404	1212	33.0693
30	4	1920	384	1152	33.7500
32	2	700	140	420	20.0000
35	2	780	156	468	20.0000
37	2	2000	400	1200	20.0000
38	6	2400	480	1440	30.0000
40	4	1900	380	1140	32.6316
46	2	2000	400	1200	20.0000
50	2	1200	240	720	40.0000
54	2	1320	264	792	40.0000
57	4	1800	360	1080	33.3333

S4. SUPPLEMENTARY FIGURES

Figure S1 shows the topology of the modified IEEE 14-bus test system. The five thermal units are located at Buses 1, 2, 3, 6, and 8, the 100 MW wind farm is connected at Bus 3, and the renewable-share tracing requirement is imposed at Bus 4. Branch 4–5 is removed compared with the standard MATPOWER `case14`, which makes the share-tracing results easier to interpret.

Figure S2 compares the network predictions with the ground-truth tracing results on the held-out samples, separated into low-wind and high-wind conditions. Each point corresponds to one sample; the closer the points lie to the diagonal line, the more accurate the neural-network surrogate. The parity behavior under both regimes confirms that the surrogate remains accurate across the operating range covered by the training data.

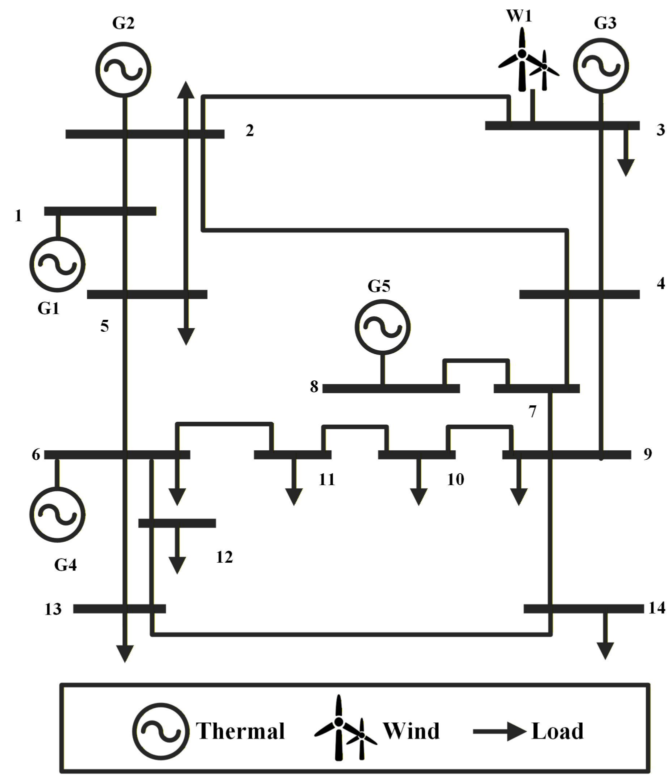

Fig. S1. Topology of the modified IEEE 14-bus test system.

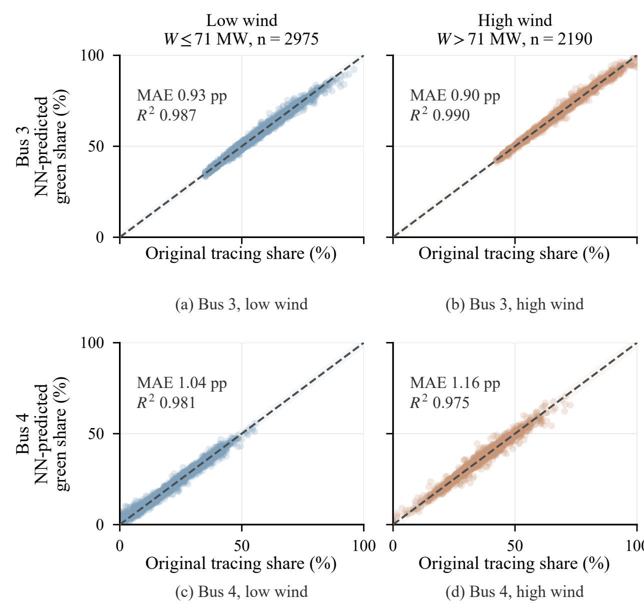

Fig. S2. Held-out parity comparison under low- and high-wind conditions.